

Adil NAS

Mersin, Türkiye • adilnasceng@gmail.com • +90 544 521 9366

GitHub: github.com/Adilnasceng LinkedIn: linkedin.com/in/adilnas Web: adilnas.dev

Summary

Honors graduate and M.Sc. student in Computer Engineering at Sivas Cumhuriyet University, specializing in robotics, autonomous systems and software development. Led my university team to 2nd place in Turkey at Teknofest two years in a row (2025 autonomous robot, 2026 autonomous forklift AGV). Completed AI and machine-learning training at Samsung Innovation Campus. Experienced in embedded systems and full-stack development, with publications on autonomous robotics, multimodal AI and software defect prediction.

Publications

NAS, A.; Kekül, H.; Yüksek, A. G. "Comparative Analysis of Data Balancing and Dimensionality Reduction Strategies in Software Defect Prediction." ICIAS 2026 (International Conference). [View Paper](#)

NAS, A.; Mandacı, E. D.; Yüksek, A. G. "Multimodal AI Systems for Complex Engineering Problems." Advanced AI Architectures for Engineering Innovation (Book Chapter, Chapter VIII, p. 185). [View Paper](#)

NAS, A.; Mandacı, E. D.; Yüksek, A. G. "IoT-Enhanced MIS Architectures for Autonomous Robotic Systems." Information Technologies in Managerial Decision-Making: Foundations and Application (Book Chapter), 2025. [DOI](#)

Education

Cumhuriyet University — Computer Engineering, M.Sc.

GPA: 3.75 / 4.00

Sivas

Sep 2025 – Present

Thesis: *Autonomous Mobile Robot Navigation System Based on Visual, Auditory and Spatial Sensor Data Fusion*

Cumhuriyet University — Computer Engineering, B.Sc.

GPA: 3.03 / 4.00 – Honor Student

Sivas

Sep 2021 – Jun 2025

Experience

Yigido Team (Teknofest) — Robotics Team Lead

Sivas

Sep 2025 – Sep 2026

- Placed 2nd in Turkey at Teknofest 2026 Robotic Applications in Industry with an autonomous forklift AGV.
- Designed the full system across electronics, hardware and software, with a Jetson Orin Nano as the main controller and an STM32 for sensor processing; planned and delegated tasks within the team.
- Built graph-based route recording, LiDAR/encoder/IMU localization and map-aware obstacle detection.
- Implemented precise docking via QR codes and camera lane tracking, plus auto-charging on low battery.
- Built PLC communication for gate access, a robot state system (loaded, unloaded, return-home, e-stop) with audio/LED/GUI feedback, and a Wi-Fi GUI with autonomous, joystick and manual modes.

Yigido Team (Teknofest) — Robotics & AI Engineer

Sivas

Sep 2024 – Sep 2025

- Won 2nd in Turkey at Teknofest 2025 Digital Technologies in Industry (Advanced) with an autonomous robot.
- Built all ROS2 software: sensor integration, SLAM (SLAM Toolbox) and autonomous navigation (Nav2).
- Owned the software end to end; contributed to hardware, electronics and sensor fusion.

Samsung Innovation Campus — AI Intern

Remote

Jul – Nov 2025

- Completed a structured AI program covering machine-learning and deep-learning fundamentals.
- Performed data analysis, model development and evaluation in Python.
- Gained hands-on experience with LLM orchestration tools such as Ollama, CrewAI and LangChain.
- Built and presented an end-to-end AI capstone project.

C-Prot — Software Intern

Mersin

Jul – Aug 2024

- Developed file-scanning and malware-detection algorithms in C++.

- Built a project focused on verifying the digital certificate status of executable files.
- Developed a full-stack application in Tauri using Svelte and Rust.

Projects

Multi-Robot Autonomous Warehouse System — ROS2 Humble, Nav2, Gazebo, Docker

- Built a centrally managed system where 4 AGVs autonomously run pick-and-drop tasks in Gazebo.
- Implemented conflict-free path planning with Conflict-Based Search (CBS) + A* and task allocation via Hungarian/greedy algorithms, plus battery and auto-charging management.
- Designed a modular 6-package ROS2 architecture with a real-time Flask/SocketIO web dashboard.

ROS2 Differential-Drive Hardware Interface — ros2_control, C++, Nav2

- Developed a ros2_control SystemInterface serial bridge for motor/encoder control, integrated with the full Nav2 stack (SLAM, AMCL, EKF sensor fusion).
- Designed it for Arduino Mega and STM32F407 boards across ROS2 Humble/Jazzy distributions.

Autonomous Line-Following Robot — Raspberry Pi, OpenCV, PID

- Built a PiCamera line-following robot using contour-based line position/angle and PID motor control.

Security & Browser Tools — Chrome MV3, JavaScript

- Built an ad blocker (220+ rules), an HTTP-analysis DevTools extension and cookie/JWT security tools.

Technical Skills

Programming Languages: C++, Python, JavaScript, C#, Java, HTML/CSS, SQL

Robotics & AI: ROS2, Nav2, SLAM Toolbox, ros2_control, Gazebo, OpenCV, Ollama, CrewAI, LangChain

Technologies & Tools: React, Node.js, Flask, Tauri, Svelte, Docker, Linux, Git/GitHub, Arduino, Raspberry Pi, Jetson, STM32

Protocols & Interfaces: I2C, SPI, UART/USART, CAN, PWM, GPIO, PLC communication

Certifications

- Teknofest 2025 – Digital Technologies in Industry, Turkey 2nd Place Certificate · TEKNOFEST (2025)
- Samsung Innovation Campus · Samsung Electronics (2025)
- ROS2 | Advanced ROS2: Modeling, Nav2 & SLAM · Udemy (2025)
- OpenCV Bootcamp · Python for Beginners · OpenCV University (2026)
- TEI Aircraft Engines School · TEI / TUSAŞ Engine Industries (2026)
- Version Control: Git & GitHub · Database Attacks and Database Security · BTK Academy (2024)

Languages

- **English:** Intermediate (B1–B2)
- **Turkish:** Native

Soft Skills

Fast Learning & Problem Solving • Team Leadership & Collaboration • Research Skills • Tech Curiosity

Interests

Robotics, Autonomous Systems, Embedded Systems, AI, Web Technologies, Open Source

References

Available upon request.